Cyclotomic Product-Row Support:
Principal Subresultants, Fitting Loci, and Resultant Transversality

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Abstract

We sharpen an exact arithmetic support problem arising from fourth-moment reductions for arithmetic quantum cat maps. For odd prime order $K = 2m + 1$, a fixed-product decomposition reduces the non-forced Cayley-energy excess to intersections of special affine copies of the prime cyclotomic trace set. A specialization-safe maximal-real formulation associates to every normalized row ratio $r \neq 1$ two degree-m polynomials P, Q over $\mathbf{Z}[\tau]$, $\tau = \zeta_K + \zeta_K^{-1}$, such that over every split residue embedding the row-intersection number equals $\deg \gcd(P, Q)$.

We prove that the order-three event is detected by only the first three principal subresultant coefficients

$$\mathrm{psc}_0(P, Q), \quad \mathrm{psc}_1(P, Q), \quad \mathrm{psc}_2(P, Q),$$

and identify the same event intrinsically as the third-corank Fitting locus of the $2m \times 2m$ Sylvester presentation. Thus the previously used full subresultant coefficient ideal and scalar shift-resultant coordinates are different coordinate systems on one determinantal stratum.

For the natural row translation

$$\mathcal{R}_r(Z) = \mathrm{Res}(P, Q_Z) = c_0 + c_1 Z + c_2 Z^2 + \cdots,$$

we prove a transversality formula at every stratum. If $P = GA$, $Q = GB$, $d = \deg G$, then the first nonzero shift coefficient satisfies, up to the fixed resultant/subresultant sign convention,

$$c_d = \alpha_r^{-d} \, \mathrm{psc}_d(P, Q) \, \mathrm{Disc}(G) \, \mathrm{Res}(A, G) \, \mathrm{Res}(G, B).$$

Every extra factor is nonzero on the maintained squarefree split locus. Hence the shift and principal-subresultant coordinates are transverse local coordinates on the same gcd stratification.

For the inverse ratio $r = -1$, reflection halves the degree. Writing $F(X) = E(X^2) + XO(X^2)$, the inverse shift resultant factors

$$\mathrm{Res}_X(F(X), F(Z - X)) = 2^m F(Z/2) S(Z)^2, \quad S(Z) = \prod_{i<j} (Z - x_i - x_j),$$

and furthermore

$$S(Z) = \pm \, \mathrm{Res}_Y(E_Z, O_Z), \quad F(X + Z/2) = E_Z(X^2) + XO_Z(X^2).$$

At a unique common root y of E, O the translation derivative is

$$\rho'(0) = \pm \, \mathrm{Res}\left(\frac{E}{Y - y}, \frac{O}{Y - y} \right) \left(E'(y)^2 - y O'(y)^2 \right),$$

where the final factor equals $-F'(z)F'(-z)/(4y)$ for $z^2 = y$. Squarefreeness therefore forces transversality at gcd degree one; vanishing of both $\rho(0)$ and $\rho'(0)$ is exactly the degree-at-least-two event.

These results reduce the remaining pointwise arithmetic problem to common split prime divisors of small canonical Sylvester systems. They do not prove a polynomial largest-prime bound and do not prove microscopic BGS universality.

1 Fixed-degree row formulation

Let $K = 2m + 1 \geq 7$ be an odd prime, let $\zeta = \exp(2\pi i/K)$, and put $\tau = \zeta + \zeta^{-1}$. The full subgroup trace polynomial is

$$H_K(T) = (T - 2)C_K(T),$$

where C_K is the minimal polynomial of τ. For a normalized row ratio $r \neq 1$ the affine trace map has the Chebyshev form

$$Y = \alpha_r X + \beta_r, \qquad \alpha_r = U_{r-1}(\tau/2), \qquad \beta_r = -2U_{r-2}(\tau/2).$$

The two full row trace polynomials share one characteristic-zero trace corresponding to the invalid row class. Removing this trace before reduction gives fixed-degree cofactors

$$P_{K,r}(T), Q_{K,r}(T) \in \mathbf{Z}[\tau][T], \qquad \deg P_{K,r} = \deg Q_{K,r} = m.$$

At every rational prime $p \equiv 1 \pmod{K}$ and every embedding prime $\mathfrak{q} \mid p$, the leading coefficients remain nonzero and

$$I_{1,r}(\mathfrak{q}) = \deg \gcd(P_{K,r}, Q_{K,r} \bmod \mathfrak{q}). \tag{1}$$

This fixed-degree statement is the specialization safeguard. Earlier projective dehomogenizations can lose degree at the point at infinity and are therefore retained only as one-way obstructions.

2 Principal subresultants are sufficient

For two fixed-degree polynomials P, Q over a field, let $S_j(P, Q; T)$ be the standard jth subresultant polynomial and let

$$\mathrm{psc}_j(P, Q) = [T^j]S_j(P, Q; T).$$

We use the standard subresultant theorem [1]: if $g = \deg \gcd(P, Q)$, then

$$S_0 = \cdots = S_{g-1} = 0, \qquad S_g = c \gcd(P, Q), \qquad c \neq 0.$$

Lemma 2.1 (Principal coefficients detect gcd degree)**.** *For every $d \geq 1$,*

$$\deg \gcd(P, Q) \geq d \iff \mathrm{psc}_0(P, Q) = \cdots = \mathrm{psc}_{d-1}(P, Q) = 0.$$

Proof. If the gcd degree is at least d, the first d subresultants vanish. If the gcd degree is $g < d$, the first nonzero subresultant is $c \gcd(P, Q)$ of degree exactly g, so its T^g coefficient is nonzero. □

Apply this with $d = 3$ to the fixed-degree row cofactors. Define

$$\pi_j = \mathrm{psc}_j(P_{K,r}, Q_{K,r}), \qquad j = 0, 1, 2,$$

and

$$J^{\mathrm{psc}}_{K,r} = (\pi_0, \pi_1, \pi_2) \subset \mathbf{Z}[\tau].$$

Let

$$J^{\mathrm{psc}}_{K,r} \cap \mathbf{Z} = \Pi^{(3)}_{K,r}\mathbf{Z}.$$

Theorem 2.2 (Three-scalar exact split support)**.** *For every split rational prime* $p \equiv 1 \pmod{K}$*,*

$$p \mid \Pi^{(3)}_{K,r} \iff \textit{some } \mathfrak{q} \mid p \textit{ has } I_{1,r}(\mathfrak{q}) \geq 3.$$

Proof. Combine (1) with Lemma 2.1. A rational prime divides the contraction of an ideal precisely when the ideal is contained in some prime ideal above it. □

Thus the large full-coefficient subresultant support and the three principal coefficients have the same split rational-prime support. Their valuations need not agree. Exact Smith-normal-form controls include

(K, r)	$\Pi^{(3)}_{K,r}$	prime-to-$2K$ part
$(7, 2)$	7	1
$(11, 7)$	11^2	1
$(13, 3)$	13^3	1
$(13, 12)$	$2^4\,5^3\,13^3$	5^3
$(17, 10)$	$2^2\,17^5\,103$	$103.$

At $(K, r) = (13, 12)$ the independently maintained full/shift support has prime-to-$2K$ valuation 5^2 rather than 5^3. This is an explicit negative control against identifying the support integers themselves.

3 The intrinsic Sylvester/Fitting locus

Let

$$\mathcal{S}_{P,Q} : k[T]_{<m} \oplus k[T]_{<m} \longrightarrow k[T]_{<2m}, \qquad (U, V) \mapsto UP + VQ.$$

In monomial bases this is the $2m \times 2m$ Sylvester matrix.

Theorem 3.1 (Sylvester nullity)**.** *For same-degree nonzero polynomials over a field,*

$$\dim \ker \mathcal{S}_{P,Q} = \deg \gcd(P, Q).$$

Proof. Write $P = GP_1$, $Q = GQ_1$ with $(P_1, Q_1) = 1$ and $g = \deg G$. A kernel vector satisfies $UP_1 = -VQ_1$, so $U = WQ_1$, $V = -WP_1$. The degree bounds are exactly $\deg W < g$. □

Consequently

$$\deg \gcd(P, Q) \geq 3 \iff \operatorname{rank} \mathcal{S}_{P,Q} \leq 2m - 3.$$

Let $I_{2m-2}(\mathcal{S})$ be the ideal generated by all $(2m-2) \times (2m-2)$ minors. Then

$$\deg \gcd(P, Q) \geq 3 \iff I_{2m-2}(\mathcal{S}) = 0$$

after specialization to a field. Equivalently, the $(2m-2)$nd compound matrix vanishes. If the gcd degree is exactly two, this compound matrix has rank one.

Thus the order-three row support is intrinsically the third-corank Fitting locus of a cyclotomic Sylvester presentation. The PSC triple is a small canonical coordinate system cutting out the same field-valued support, while the all-minor Fitting ideal exposes exterior-power and determinantal identities.

4 Shift-resultant transversality

Let k be a field, let f, g be monic degree-n polynomials, and suppose

$$f = GA, \qquad g = GB,$$

where G is monic squarefree of degree d and $(A, B) = 1$. Consider a degree-preserving deformation

$$g_Z = g + Z\psi + O(Z^2)$$

and assume $\psi(y) \neq 0$ at every root y of G. Put

$$R(Z) = \operatorname{Res}(f, g_Z).$$

Theorem 4.1 (First nonzero shift coefficient)**.** *One has* $\operatorname{ord}_{Z=0} R = d$*, and, up to the fixed resultant/subresultant sign convention,*

$$[Z^d]R(Z) = \operatorname{psc}_d(f, g)\ \operatorname{Res}(A, G) \prod_{G(y)=0} \psi(y).$$

Proof. The roots of f coming from G each contribute one factor $Z\psi(y)+O(Z^2)$. The remaining roots, those of A, contribute

$$\operatorname{Res}(A, g) = \operatorname{Res}(A, G) \operatorname{Res}(A, B).$$

For monic f, g, G the first nonzero subresultant is $\pm \operatorname{Res}(A, B)G$, so its principal coefficient is $\pm \operatorname{Res}(A, B)$. □

For the row shift, normalize the target polynomial to be monic and write

$$q_Z(T) = \alpha_r^{-m} R_r(\alpha_r T + \beta_r + Z).$$

Then

$$\psi(T) = \alpha_r^{-m} R_r'(\alpha_r T + \beta_r), \qquad q_0'(y) = \alpha_r \psi(y).$$

On the maintained split locus the target cofactor is squarefree and $\alpha_r \neq 0$, so every $\psi(y)$ is nonzero.

If $G = \gcd(P, q_0)$, the product in Theorem 4.1 can be evaluated further:

$$\prod_{G(y)=0} \psi(y) = \pm \alpha_r^{-d} \operatorname{Disc}(G) \operatorname{Res}(G, B).$$

Hence

$$[Z^d] \operatorname{Res}(P, q_Z) = \pm \alpha_r^{-d} \operatorname{psc}_d(P, q_0) \operatorname{Disc}(G) \operatorname{Res}(A, G) \operatorname{Res}(G, B). \tag{2}$$

Every multiplier relating the first nonzero shift coordinate to the first nonzero PSC is nonzero in the residue field. Thus the shift coefficients and PSCs are transverse coordinate systems on the same gcd stratification.

In particular,

$\deg \gcd$	PSC coordinates	shift coordinates
0	$\pi_0 \neq 0$	$c_0 \neq 0$
1	$\pi_0 = 0,\ \pi_1 \neq 0$	$c_0 = 0,\ c_1 \neq 0$
2	$\pi_0 = \pi_1 = 0,\ \pi_2 \neq 0$	$c_0 = c_1 = 0,\ c_2 \neq 0$
≥ 3	$\pi_0 = \pi_1 = \pi_2 = 0$	$c_0 = c_1 = c_2 = 0.$

5 Inverse pair sums and half-degree support

For $r = -1$, reflection gives

$$Q(T) = -P(2\tau - T).$$

Set

$$F(X) = P(\tau + X) = E(X^2) + XO(X^2).$$

Then over every split embedding prime

$$I_{-1} = 2 \deg \gcd(E, O).$$

Let $x_1, \ldots, x_m$ be the roots of F and put

$$S(Z) = \prod_{i<j} (Z - x_i - x_j).$$

Theorem 5.1 (Inverse pair-sum square)**.**

$$\mathrm{Res}_X(F(X), F(Z - X)) = 2^m F(Z/2) S(Z)^2.$$

At split primes $2^m F(0)$ is a unit, hence

$$I_{-1} = 2 \operatorname{ord}_{Z=0} S, \qquad I_{-1} \geq 3 \iff S(0) = S'(0) = 0.$$

The pair-sum polynomial itself is a half-degree resultant. If

$$F(X + Z/2) = E_Z(X^2) + XO_Z(X^2),$$

then

$$S(Z) = \pm \mathrm{Res}_Y(E_Z, O_Z). \tag{3}$$

To first order,

$$\partial_Z E_Z|_0 = \frac{O + 2YO'}{2}, \qquad \partial_Z O_Z|_0 = E'.$$

Thus inverse order-three support is a two-scalar problem: a half-degree resultant and its translation derivative.

6 Inverse resultant transversality

We record a local resultant derivative identity in a form independent of the cyclotomic application.

Theorem 6.1 (Simple-common-root derivative)**.** *Let*

$$f = (Y - y)A, \qquad g = (Y - y)B, \qquad (A, B) = 1,$$

and assume y is simple in both f and g. For $f_t = f + t\phi$, $g_t = g + t\psi$,

$$\frac{d}{dt} \mathrm{Res}(f_t, g_t)\Big|_{t=0} = (-1)^{\deg f - 1} \mathrm{Res}(A, B)\big(f'(y)\psi(y) - g'(y)\phi(y)\big).$$

Proof. Let $y(t)$ be the perturbed root of f_t with $y(0) = y$. Then $y'(0) = -\phi(y)/f'(y)$ and

$$g_t(y(t)) = t\left(\psi(y) - \frac{g'(y)\phi(y)}{f'(y)}\right) + O(t^2).$$

The other roots of f contribute the nonzero product $\mathrm{Res}(A, g) = \mathrm{Res}(A, Y - y)\,\mathrm{Res}(A, B)$, and $\mathrm{Res}(A, Y - y) = (-1)^{\deg f - 1} f'(y)$. □

Apply this to

$$\rho(Z) = \operatorname{Res}(E_Z, O_Z).$$

At a unique common root y of E, O,

$$\rho'(0) = \pm \operatorname{Res}\left(\frac{E}{Y-y}, \frac{O}{Y-y}\right)(E'(y)^2 - yO'(y)^2).$$

If $z^2 = y$, then direct differentiation of $F(X) = E(X^2) + XO(X^2)$ gives

$$E'(y)^2 - yO'(y)^2 = -\frac{F'(z)F'(-z)}{4y}.$$

The maintained split trace polynomial is squarefree and $F(0)$ is a unit, so this factor is nonzero whenever the gcd degree is one. Therefore

$$\deg\gcd(E, O) = 1 \Longrightarrow \rho(0) = 0, \quad \rho'(0) \neq 0,$$

while

$$\deg\gcd(E, O) \geq 2 \iff \rho(0) = \rho'(0) = 0.$$

The second scalar is therefore a genuine transversality test, not an accidental coefficient.

More generally, if a squarefree gcd $G = \prod_{j=1}^{d}(Y - y_j)$ is shared by $f = GA$, $g = GB$, then for a directional perturbation $f_t = f + t\phi$, $g_t = g + t\psi$ the first nonzero resultant coefficient is

$$[t^d] \operatorname{Res}(f_t, g_t) = \operatorname{Res}(A, G) \operatorname{Res}(A, B) \prod_{j=1}^{d} \frac{f'(y_j)\psi(y_j) - g'(y_j)\phi(y_j)}{f'(y_j)}.$$

When all local Jacobians are nonzero, the vanishing order is exactly d.

7 Arithmetic frontier

The exact order-three event now has several compatible forms:

$$\begin{aligned}
\text{all ratios:} \quad I \geq 3 \quad &\iff \pi_0 = \pi_1 = \pi_2 = 0, \\
&\iff \operatorname{corank}\operatorname{Sylv}(P, Q) \geq 3, \\
&\iff c_0 = c_1 = c_2 = 0, \\
\text{inverse ratio:} \quad I_{-1} \geq 3 \quad &\iff \deg\gcd(E, O) \geq 2, \\
&\iff \rho(0) = \rho'(0) = 0.
\end{aligned}$$

The characteristic-zero affine-intercept theorem ensures that the family does not lie on the order-three stratum before reduction. Finite exceptional primes are therefore genuine rank-drop primes of a structured cyclotomic Sylvester module.

The remaining arithmetic theorem is not a height bound for one resultant. It is common-divisor control for the small canonical systems above. A sufficient global form remains, for example,

$$P^{+}_{\text{split}}\left(\prod_{[r]} \Pi^{(3)}_{K,r}\right) \leq K^{4+o(1)}.$$

A fixed-ratio logarithmic support budget $O(K^3)$ is already known, but it does not imply this largest-prime statement.

The Fitting formulation points to concrete next attacks: identities among neighboring Sylvester minors (Desnanot–Jacobi/Dodgson), subresultant PRS or Habicht identities, Bezoutian and reduced-resultant formulations, and modular Smith/Fitting invariants. The goal is to control the common split prime support of consecutive determinantal coordinates, not their individual norms.

8 Verification boundary

The repository maintains exact symbolic and finite-field regressions for every coordinate system above. Principal-subresultant and inverse transversality gates have passed the full scientific CI together with C++20, Lean 4 and an independent TeX build. Focused row-support workflows separately exercise the new shift/PSC and Fitting formulations. Finite computations verify implementations and explicit anchors; they are not the proof of the universal statements.

Representative valuation controls are

$$\Pi'_{13,12} = 5^3, \qquad \Lambda'_{13,12} = 5^2,$$

and

$$\Theta'_{23} = 47^2 \cdot 137, \qquad \Xi'_{23} = 47 \cdot 137.$$

Thus equality of split prime support must never be silently upgraded to integer or valuation equality.

None of these row-support theorems proves microscopic BGS/GOE/COE universality. Even a completed pointwise fourth-moment theorem leaves the independent Heisenberg-scale joint-trace comparison and high-frequency tail problems.

References

[1] W. S. Brown and J. F. Traub, *On Euclid's Algorithm and the Theory of Subresultants*, J. ACM **18** (1971), 505–514, doi:10.1145/321662.321665.

[2] H. Bass, *Generators and Relations for Cyclotomic Units*, Nagoya Math. J. **27** (1966), 401–407, doi:10.1017/S0027763000026258.

[3] H. Fu, *Projective Equivalence for the Roots of Unity*, J. Algebraic Combin. **56** (2022), 823–871, arXiv:1905.01510.